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كلية علوم الحاسب وتقنية المعلومات
College of Computer Sciences & Information Technology

GLOBALTECH CORPORATION

REPORT

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Introduction

1.1 Problem Statement: GlobalTech Corporation current network infrastructure is outdated and cannot support the growing needs of the organization.

The current network at **GlobalTech Corporation** is very old and has many problems that stop the company from growing.

First, there is a **single point of failure** because the network has no redundancy. This means if one router or switch breaks, the whole company stops working. Second, the network cannot be **scaled** or expanded easily for new employees. Also, there are **no VLANs**, so every department (like HR and Sales) uses the same network. This is bad for performance and security. Finally, the network is not safe because there is **no firewall or VPN** to protect our private data or allow employees to work safely from home.

1.2 User Requirements

Include needs for reliable Wi-Fi, segmented traffic, guest access, and remote access. Reliable Wi-Fi this is mean everyone needs a strong wireless signal in all offices.

Department Separation: We must use VLANs to keep each department's traffic separate for better security.
Guest Access visitors: Should have their own internet connection that is separate from the company.
Remote Access: Employees need a secure way (SSH) to log in and work from outside the office.
High Availability: The servers must stay online all the time without any downtime.

1.3 Requirement Specifications: Outline key requirements such as scalability, high availability, and support for critical applications.

Scalability we want a network that can grow with the company. To achieve this, we chose high quality equipment like Multilayer Switches and Routers that make it easy to add or remove new employee devices without changing the whole setup.
High Availability: We need the network to work 24/7 without stopping. To achieve this, we used two or more routers and core switches.
Support for Critical Applications: the company has many important services like Web, Email, and Database servers.
We also made sure that services like DNS and DHCP are always available so that every device can get an IP and find the company's website easily.

Methodology

2.1 Data Collection: Collect information about the existing network, including bandwidth usage, traffic patterns, and employee needs.
To create the best network for the company, we needed to gather some information about the current network and how employees use it. We mean gathering information like how many people are using the network, in each department, which services they use such as mail, web, and database. We also need to look at the weak Wi-Fi coverage issues, and the only one router serving whole company, The IP range given for this project is **210.20.130.0/24**., we also take this into consideration, the gathering of the information helped us crate a better network

2.2 Data Analysis: Analyze the current network performance and identify bottlenecks and vulnerabilities.
After we collected the information we mentioned before, we analyzed the current network that we need to improve and found that the network has no separation between departments and that will lead to traffic and mix and slow everything down. When the Wi-Fi connection is weak in many areas, the employees will face problem while trying to connect. A whole company is depending on only one router, this is a single point failure that's mean when it goes down the whole company is off working, also the company current network does NOT support future growth, that's mean the company will not be able to expand and cannot hire any new employees which they need to do to grow their company. After all that we indeed sure that the company needs to redesign and improve performance, security and scalability.

2.3 Topology Design: Develop a detailed network topology that segments the network by department and provides redundancy.
For the current network, we designed a topology that's separates each department using VLANs to avoid mixed traffic and for better security. A core switch will be used, and several access switches will connect to it, and each department as well will be connected to the access switch. We will use layer switch to route between the VLANs. All the servers like DNS, mail, web, database, in their own server VLAN to keep them secure from the rest of the network. About the Wi-Fi issues, we will put multiple wireless routers across the building to improve coverage. We also will link switches so that if one goes down, the network will be still available. We also will make sure that the company will be able to expand in the future.

3 **2.4 Data Flow**: Describe how data will flow through the network, ensuring that critical services have priority and are protected from external threats.

In our new network, the data will move through the system based on the VLAN structure and routing design and as we said each department will use its own VLAN, and if they wanted to contact another VLAN the data will go to layer switch or router to be routed correctly, When a user need to access servers such as web, mail, database, or DNS their traffic will go to the server VLAN to keep it secure and separated from normal users. Also, about the firewall, when traffic goes to the internet it will pass through the firewall to check security rules. After that it will go to the router and then toward the ISP. We install multiple routers as an access point, so wireless users will connect these users to the access points which will forward the data to the nearest access switch and follow the same routing rules as wired devices. We made sure that the design is controlled, secure and send everything to it right place.

Outcomes

3.1 Network Topology: Present a logical network diagram that illustrates the new network design, including VLANs, IP addressing, and routing protocols.

In the final network, topology for the company is built using a core switch as the center of the design. That's mean all the departments are connected to access switches, then these access switches will connect back to the core switch. A layer 3 switch (A device works like a switch and a router) or a router will do the routing between the VLANs and make sure that the traffic between departments is flowing right. How will we improve the speed and security of main servers such as DNS, mail, web, and database? By placing them in their own VLAN and connected directly to the core switch. Wireless access points will be connected to the access switches to get full Wi-Fi coverage for all users in the building. A firewall is installed before the routers so that all incoming and outgoing traffic will be controlled and checked. Also, redundant links between switches in case one connection fails the network will still be working, OSPF was selected as the interior gateway protocol (IGP). This choice was made because OSPF is an open-standard link-state protocol, ensuring that GlobalTech is not locked into a single hardware vendor. Furthermore, OSPF's area-based hierarchical structure supports the company's growth plans by localizing routing updates and reducing CPU overhead on the core switches (Catalyst 9600).

3.2 Equipment Selection: Provide a detailed list of network equipment, including switches, routers, firewalls, and access points, with justification for each selection.

For the new network, we selected the equipment based on the company's needs in terms of security, performance, and future growth. We chose a core switch because it can take or can handle most of the traffic and connect all access switches to each other. We selected access switches because it will connect end devices like computers, printers and wireless access points. For the route between VLANs, we selected a router, so the traffic can move between departments fast and reliably. We installed a Firewall to protect the network and filter any harmful traffic going in or out. For the full Wi-Fi coverage for all users in office, we chose wireless access points. All these devices we named support VLANs, redundancy, and enough ports for future expansion.

Device Category	**Device Name in Packet Tracer**	**Quantity**
Core Routing	Catalyst 8500 Series Edge Platforms C8500-12X4QC	2
Core Switching	Cisco Catalyst 9600 Series Switches	2
Access Switching	Cisco Business 250 Series Smart Switches	7
Servers	Server-PT	**9**
Wireless Routers	Cisco Wireless 9179F AP	**5**
Firewall / Security	Cisco FPR9K-NM-8X10G Firewall	**2**
End Devices	P / Laptop	**~15**
Other Devices	Smartphone	**2**

3.3 IP Addressing: Show how the IP range has been subnetted to allocate addresses to different departments and services.

Department	Users	Network Address	Subnet Mask	Usable IP Range	Default Gateway
Software Dev	60	210.20.130.0/26	255.255.255.192	.2 - .62	210.20.130.1
IT Support	40	210.20.130.64/26	255.255.255.192	.66 -.126	210.20.130.65
Human Res	10	210.20.130.128/27	255.255.255.224	.130 - .158	210.20.130.129
Marketing	10	210.20.130.160/27	255.255.255.224	.162 - .190	210.20.130.161
Sales	20	210.20.130.176/28	255.255.255.240	.198 - .206	210.20.130.177
Finance	10	210.20.130.192/28	255.255.255.240	.194 - 206	210.20.130.193

3.4 Cost Estimation: Provide an estimated cost of the new network, including equipment, installation, and maintenance costs.

Item Category	Equipment Model	Quantity	Total (SAR)
Core Routing	Catalyst 8500 Series Edge Platforms C8500-12X4QC	2	1,393,482.22
Core Switching	Cisco Catalyst 9600 Series Switches	2	210,044.8
Access Switching	Cisco Business 250 Series Smart Switches	7	6,170.08
Network Security	Cisco FPR9K-NM-8X10G Firewall	2	204,673.66
Wireless Solution	Cisco Wireless 9179F AP	5	12,190.1
Server Farm	Dell PowerEdge R640	3	185,664.6
Infrastructure	Rack, Cat9 Cables, Patch Panels	1	160,000 - 220,000

Total Project Cost: 2,172,225.46 – 2,232,225.46 SAR
*All prices from Cisco page

4 Conclusion

In conclusion, the new network design for GlobalTech Corporation solves the old network problem by using VLAN separation, better security, stronger Wi-Fi coverage, and redundant links. And about the services such as DNS, mail, web, and database, we made sure that the network design supports all these requirements and keeps the servers protected in their own VLAN. Using a core switch, access switches, and a layer 3 device for routing will make the network smoother and more manageable. The new network is flexible enough for future expansion and add more users to the departments. In the end, we see that the new network will improve performance, security, and reliability for all users (employees).

5 Recommendations

For future improvements, the company could upgrade the network to IPv6 to support more devices as the company grows. Also, for better redundancy they could add more core switch instead of two, that will increase availability. Also, there is no harm to installing a new, improved or advanced Firewall to improve security. As much as users are being added as much, you can add more wireless access points to maintain strong Wi-Fi coverage. They also should always make sure or keep an eye on how the traffic is flowing to keep the network performing well.

6 References

Network Design Course Slides
Cisco Packet Tracer application

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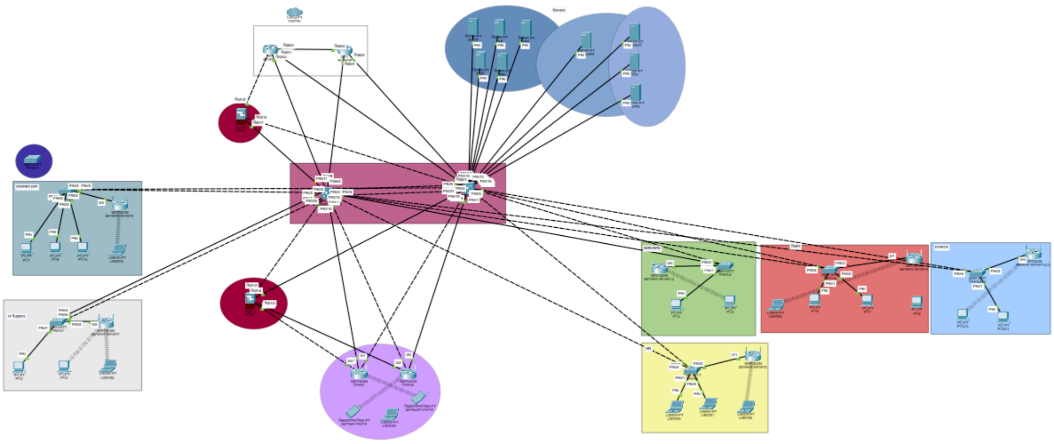
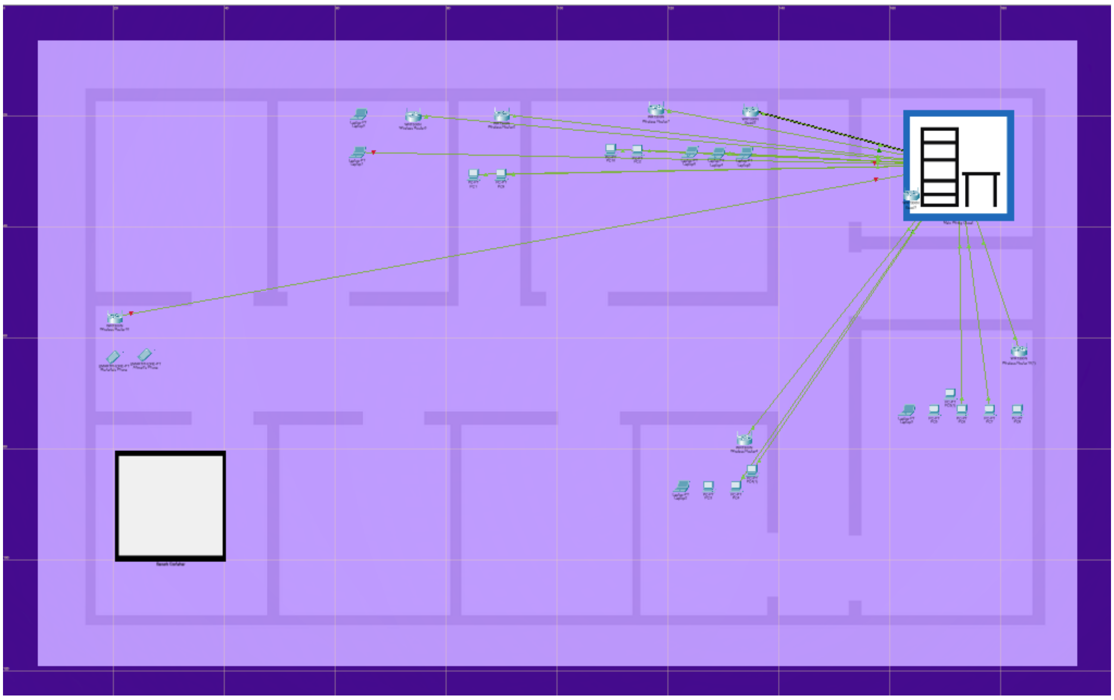
Packet Tracer Implementation

Logical Design of the new Network:



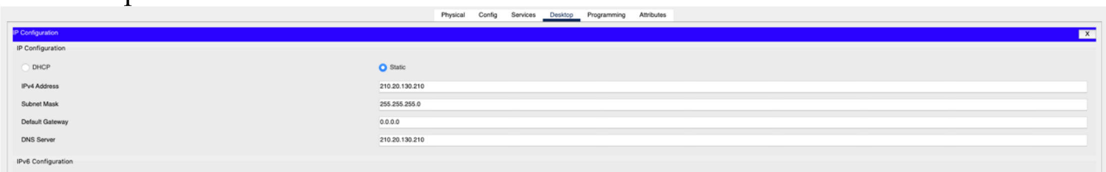
Physical Desing of the new Network:



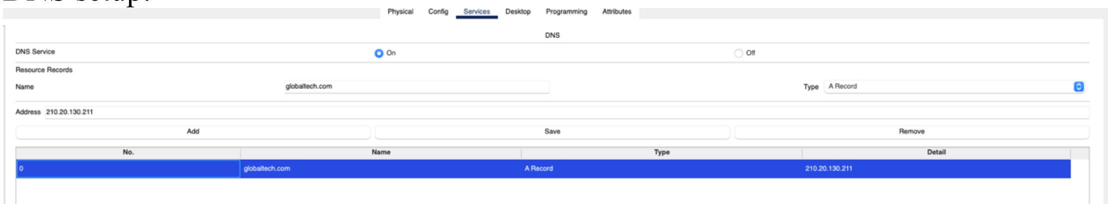
Servers:

1. Web&DNS

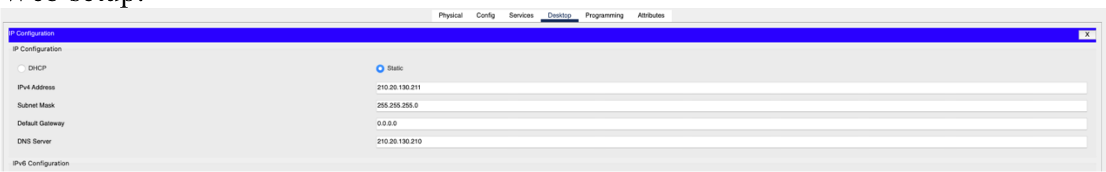
DNS setup:



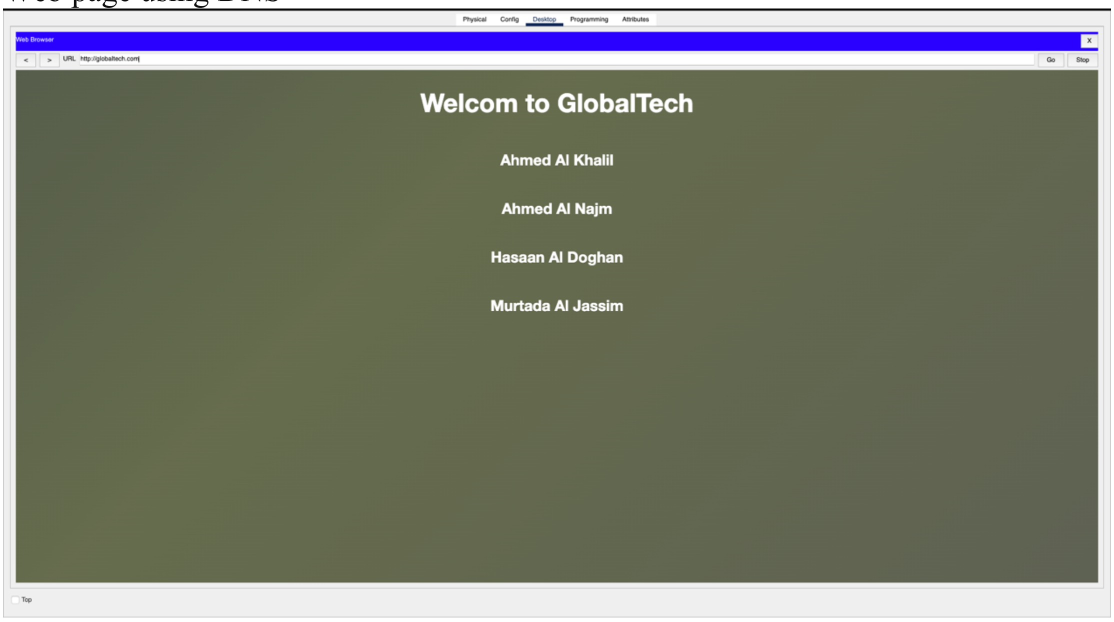
DNS setup:



Web setup:

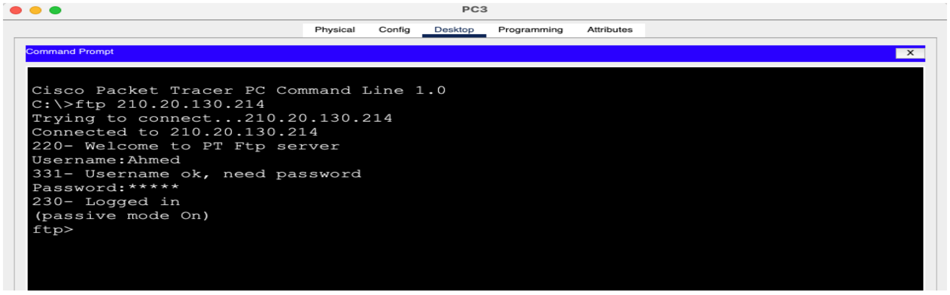


Web page using DNS

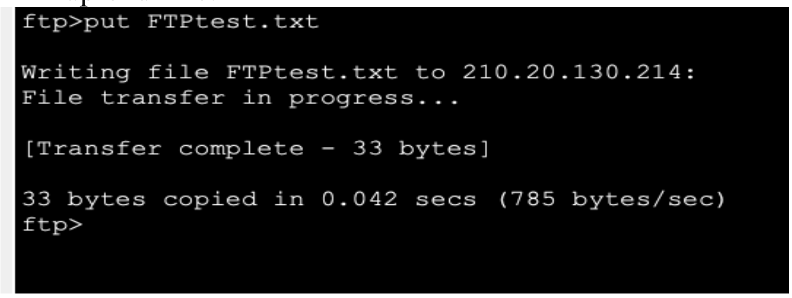


2. FTP

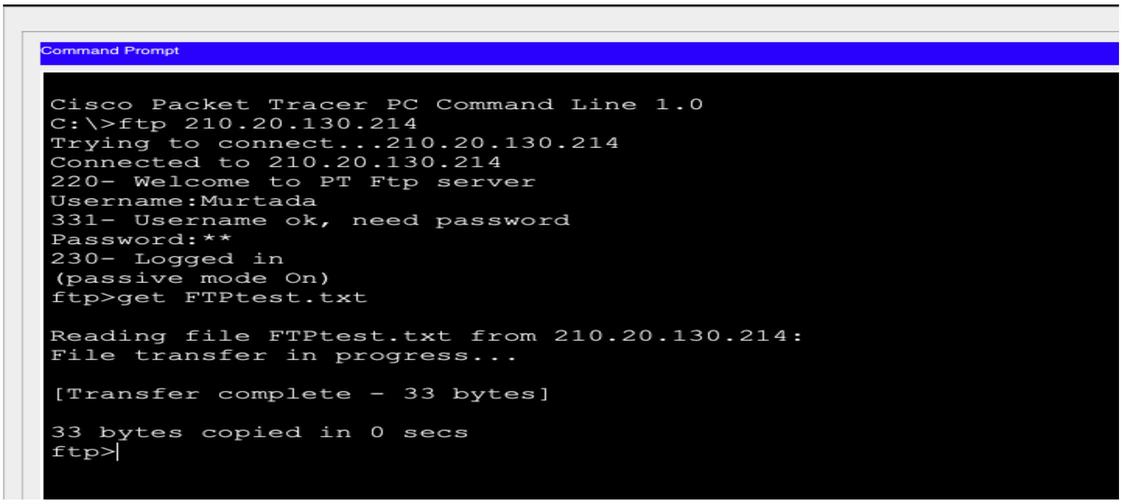
FTP Connection



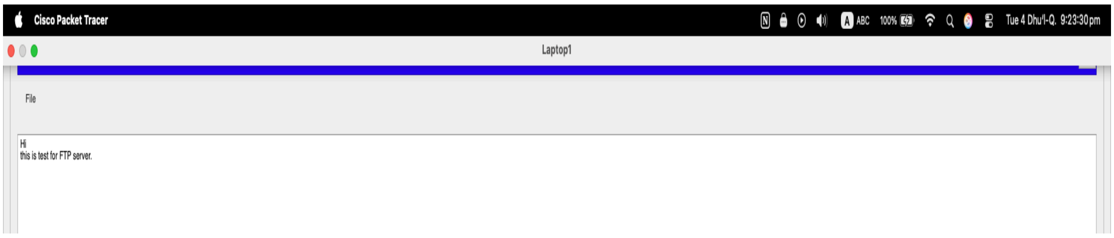
FTP upload File:



Transfer FTP

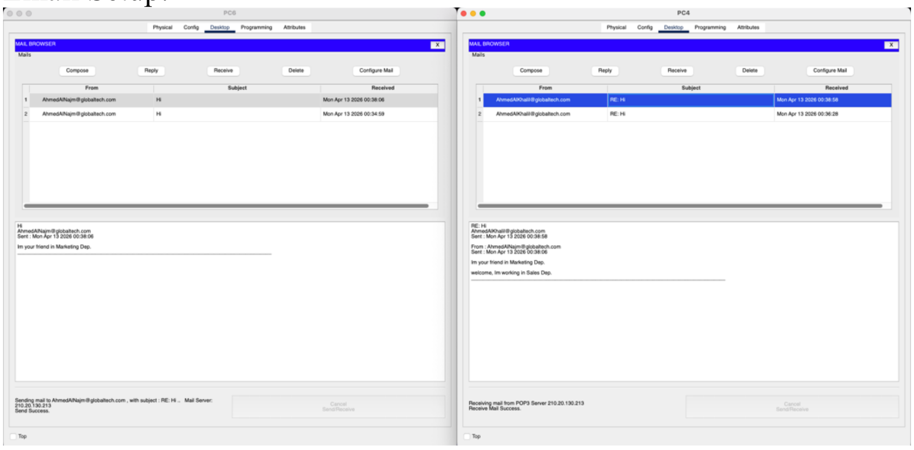


Download FTP

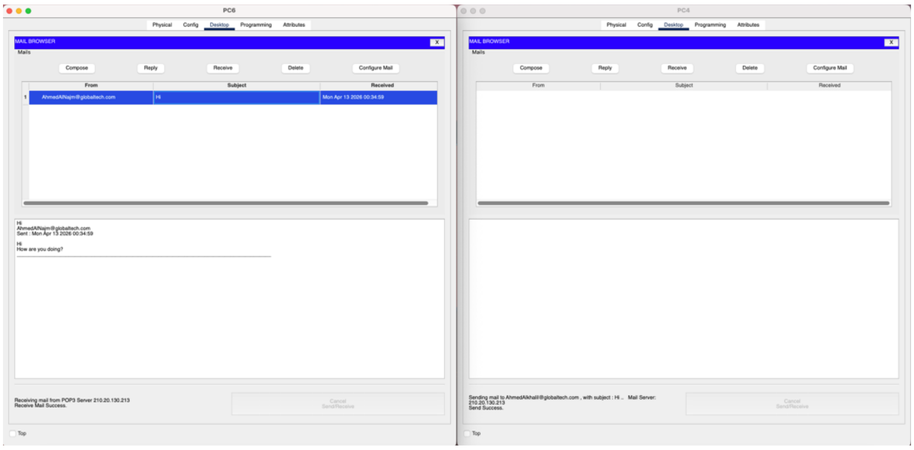


3. Email

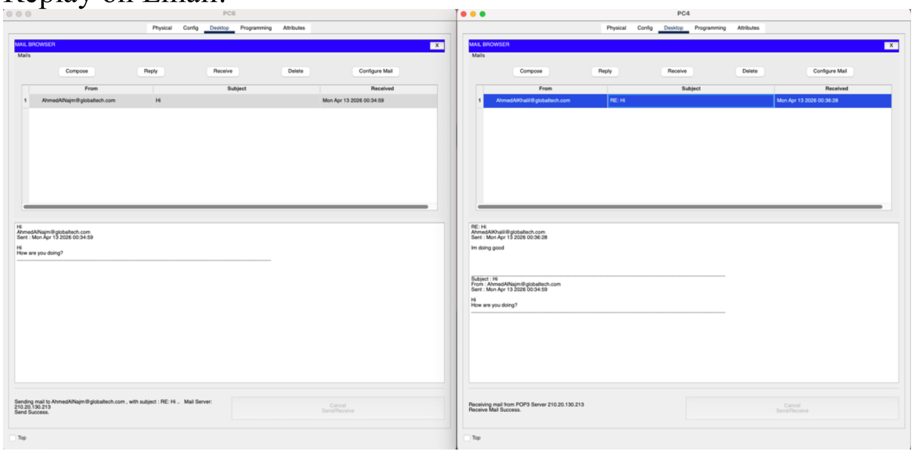
Email Setup:



Send an Email:

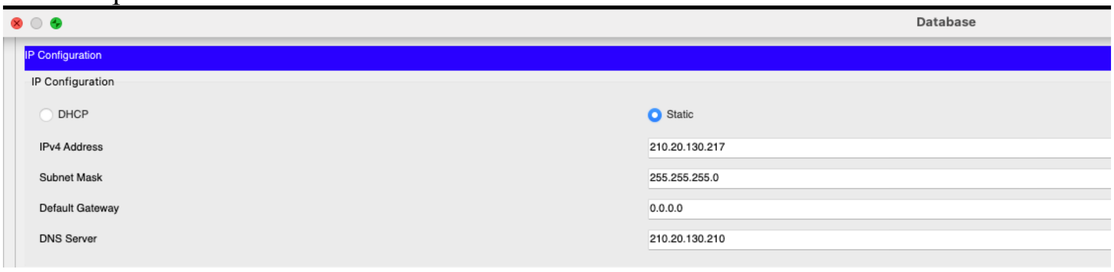


Replay on Email:



4. DataBase

DB Setup:



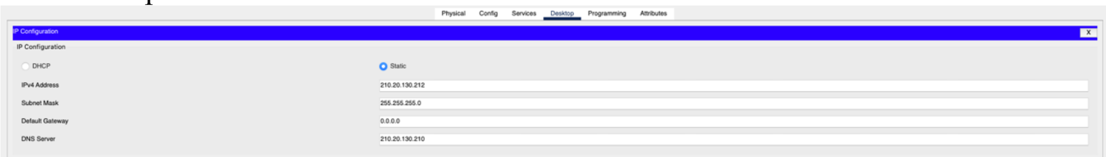
Ping:

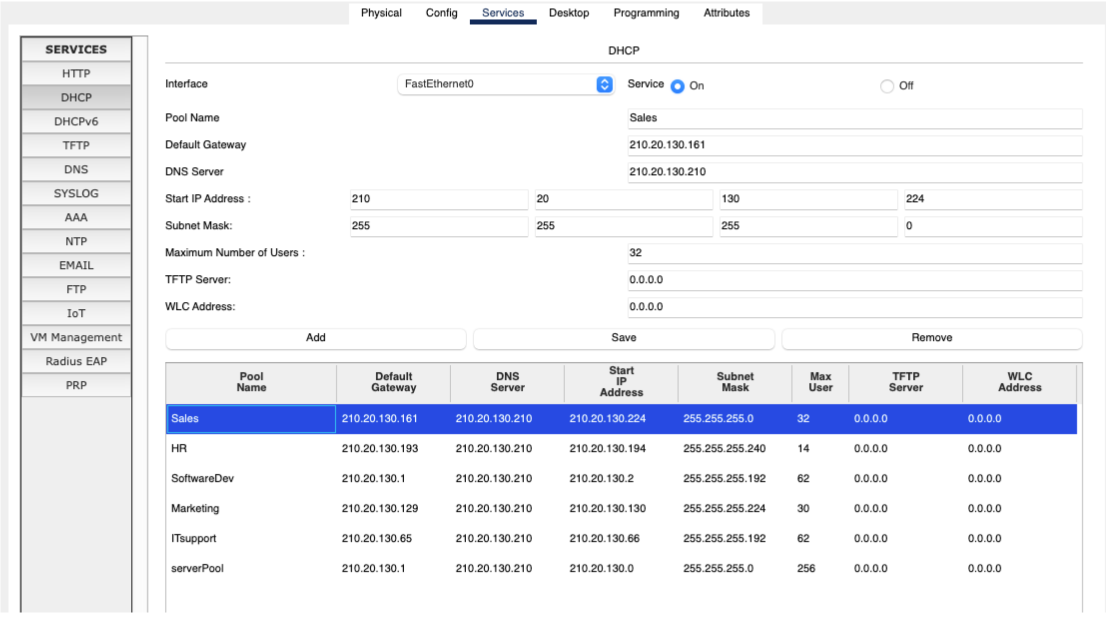
Command Prompt

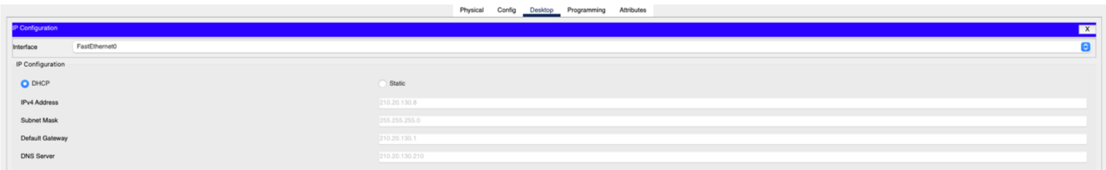
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 210.20.130.217

Pinging 210.20.130.217 with 32 bytes of data:

Reply from 210.20.130.217: bytes=32 time<1ms TTL=128
Reply from 210.20.130.217: bytes=32 time<1ms TTL=128
Reply from 210.20.130.217: bytes=32 time=1ms TTL=128
Reply from 210.20.130.217: bytes=32 time<1ms TTL=128

Ping statistics for 210.20.130.217:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

5. DHCP

DHCP setup:







6. SSH

SSH setup:

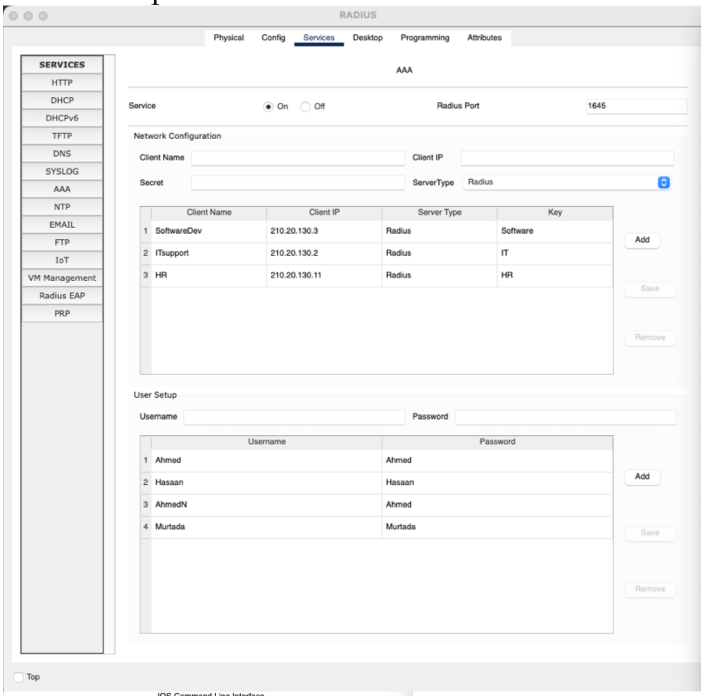
```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#en
Switch(config)#enable s
Switch(config)#enable secret 1234
Switch(config)#line 0 1
%Error: Line 1 is not in async mode
Switch(config)#line vty 0 1
Switch(config-line)#login local
Switch(config-line)#trans
Switch(config-line)#transport in
Switch(config-line)#transport input ss
Switch(config-line)#transport input ssh
Switch(config-line)#transport output ssh
Switch(config-line)#exit
Switch(config)#ip domain-
Switch(config)#ip domain-name ITsupport
Switch(config)#c
Switch(config)#cr
Switch(config)#crypto k
Switch(config)#crypto key g
Switch(config)#crypto key generate r
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#
Switch(config)#host
Switch(config)#hostname ITsupport
ITsupport(config)#crypto key generate rsa
The name for the keys will be: ITsupport.ITsupport
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

ITsupport(config)#us
*Mar 3 17:9:32.303: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 3 17:9:32.303: %SSH-5-ENABLED: SSH 1.5 has been enabled
ITsupport(config)#username AhmedAlkhalil s
ITsupport(config)#username AhmedAlkhalil secret 123
ITsupport(config)#
```

```
ITsupport(config)#line vty 0 15
*Mar 3 18:38:42.825: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

```
C:\>ping 210.20.130.216

Pinging 210.20.130.216 with 32 bytes of data:

Reply from 210.20.130.216: bytes=32 time<1ms TTL=128
Reply from 210.20.130.216: bytes=32 time<1ms TTL=128
Reply from 210.20.130.216: bytes=32 time<1ms TTL=128
Reply from 210.20.130.216: bytes=32 time<1ms TTL=128

Ping statistics for 210.20.130.216:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

7. Radius

Radius setup:



IOS Command Line Interface

```
Router>
Router>
Router>en
Router>enable
Router#config
Router#configure
Configuring from terminal, memory, or network
[terminal]?
Enter configuration commands, one per line.
End with CNTL/Z.
Router(config)#aaa new-mo
Router(config)#aaa new-model
Router(config)#rad
Router(config)#radius-server
% Incomplete command.
Router(config)#radius-server host
210.20.130.18
Router(config)#radius-server host
210.20.130.18 key Software
Router(config)#aaa athu
Router(config)#aaa ath
Router(config)#aaa auth
Router(config)#aaa authentication login
% Incomplete command.
Router(config)#aaa authentication login AAA
group radius
Router(config)#line vty 0 -15
                               ^
% Invalid input detected at '^' marker.

Router(config)#line vty 0 5
Router(config-line)#login authentication AAA
Router(config-line)#exit
Router(config)#
```

Copy Paste

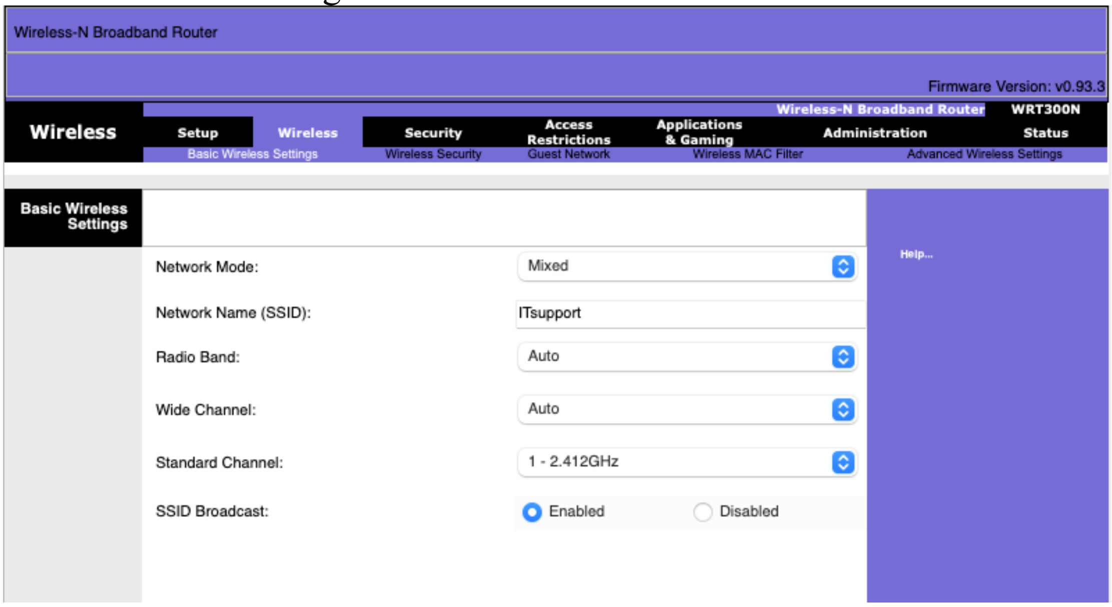
```
C:\>ping 210.20.130.218

Pinging 210.20.130.218 with 32 bytes of data:

Reply from 210.20.130.218: bytes=32 time<1ms TTL=128
Reply from 210.20.130.218: bytes=32 time=1ms TTL=128
Reply from 210.20.130.218: bytes=32 time<1ms TTL=128
Reply from 210.20.130.218: bytes=32 time<1ms TTL=128

Ping statistics for 210.20.130.218:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

8. VPN

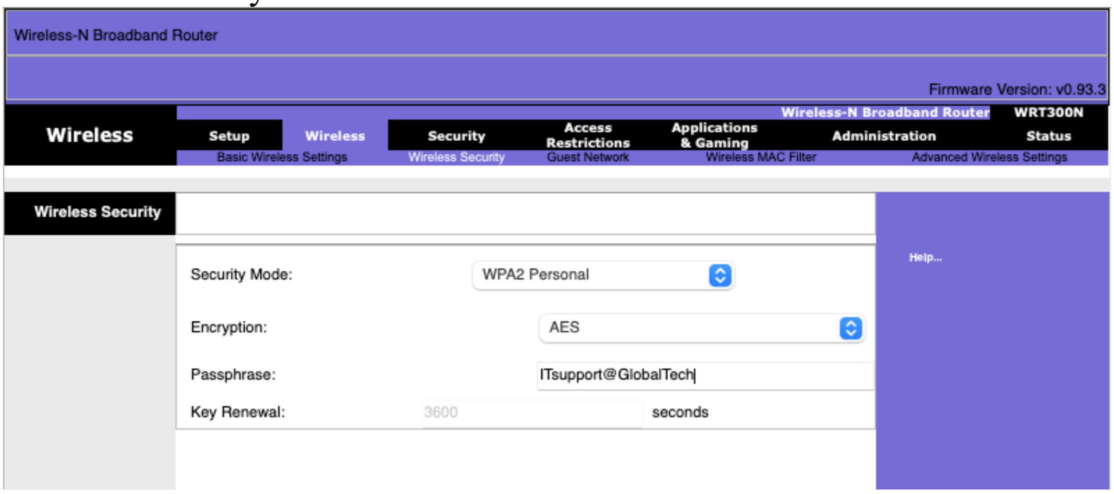
VPN ping:

```
C:\>ping 210.20.130.215

Pinging 210.20.130.215 with 32 bytes of data:

Reply from 210.20.130.215: bytes=32 time<1ms TTL=128
Reply from 210.20.130.215: bytes=32 time<1ms TTL=128
Reply from 210.20.130.215: bytes=32 time<1ms TTL=128
Reply from 210.20.130.215: bytes=32 time<1ms TTL=128

Ping statistics for 210.20.130.215:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

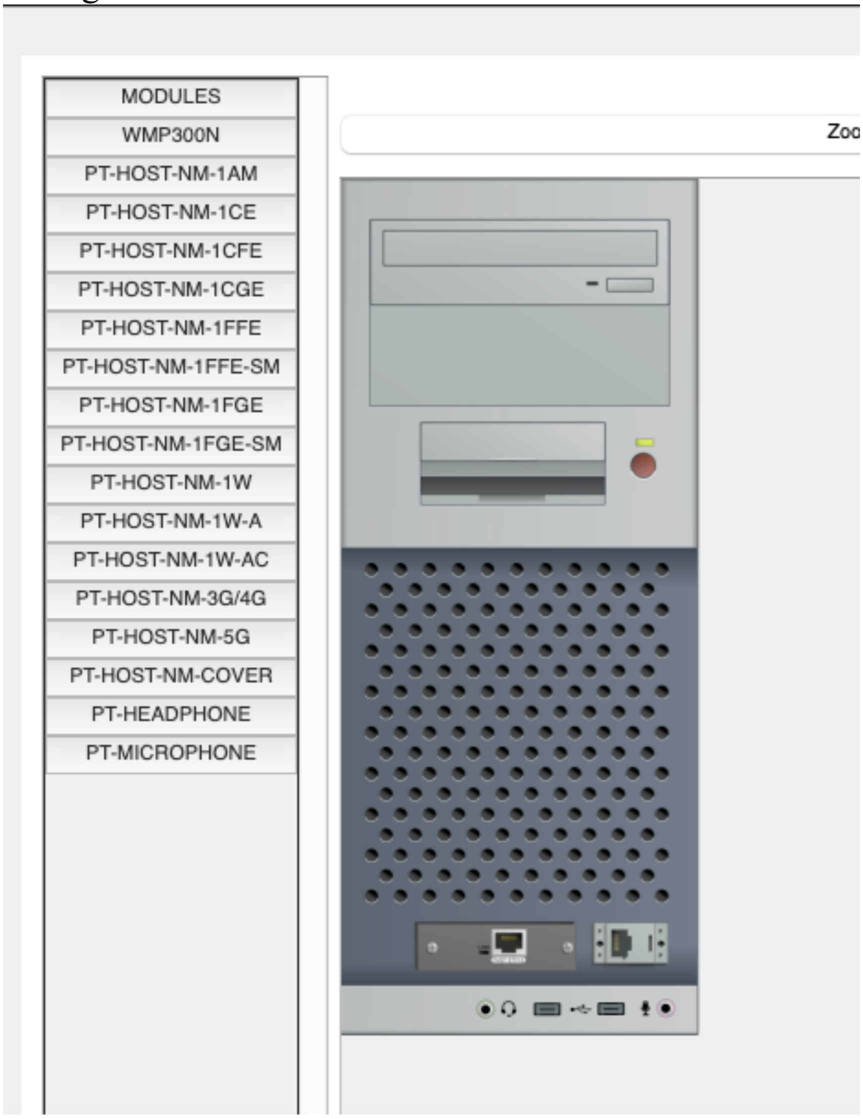
Wireless Connections
Basic Wireless Settings:



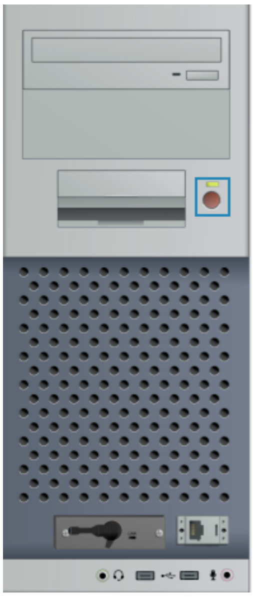
Wireless Security:



Change Network Card:



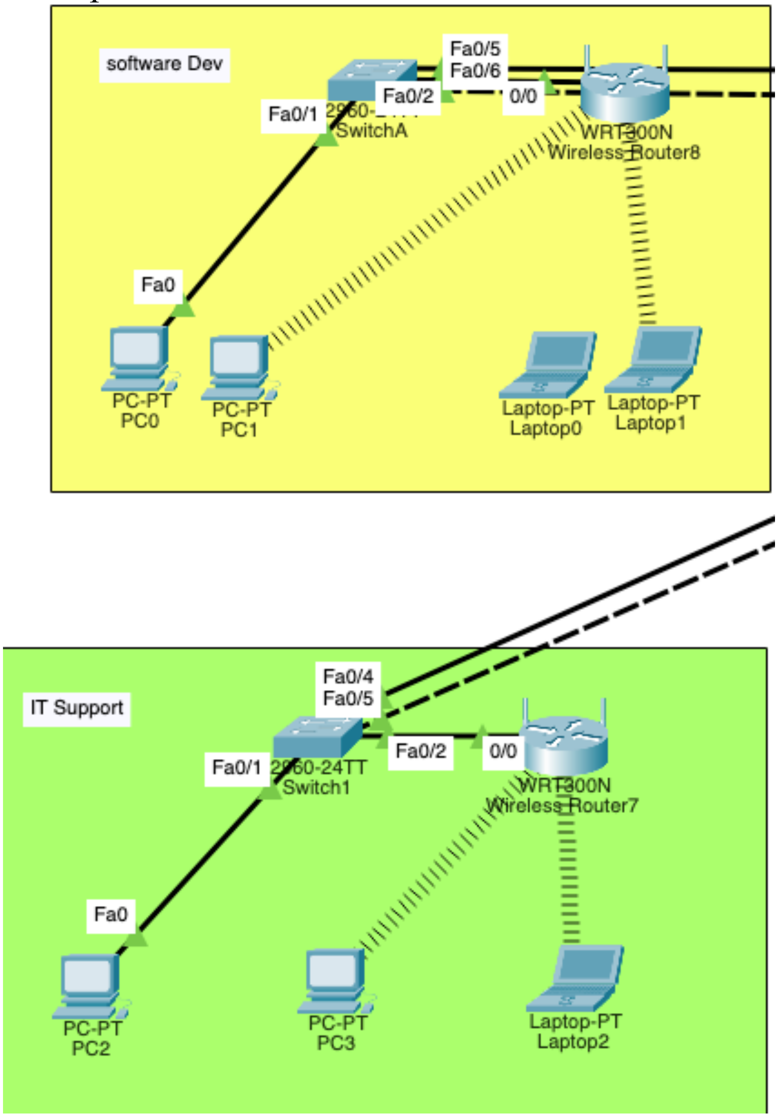
WMP300N:



Setup Connection:

Connected:

Example of Wireless Devices Connection:



Project Marking Guide

	Assignment Component	Max. Marks	Marks Obtained
Report	Format (Table of Contents, References, formatting, font, etc....)	10	
	English language (grammar and spelling)	10	
	Introduction	15	
	Contents Design, and Discussions, Packet tracer	65	
	Report mark	100 * 0.15	
Presentation	Questions and answers	15	
	Design & Usability	25	
	Testing (Various Input Conditions)	15	
	Presentation skills	15	
	Understanding / Explanation of the Design	30	
	Presentation Mark	100 * 0.05	
	Total Mark	20	
	Plagiarism report more than 25%	- 100	